

Satellite-Derived Shorelines

Joel Nicolow¹, Anna Mikkelsen², Peter Sadowski¹, Chip Fletcher²

(1)University of Hawai'i at Mānoa, Department of Information and Computer Sciences (2)University of Hawai'i at Mānoa, Department of Earth Sciences

Contact: jnicolow@hawaii.edu

Highlights

- Distance-weighted binary cross-entropy improves shoreline detection performance
- Cloud cover and other image artifacts contribute to large prediction errors, as evidenced by higher RMSE compared to MAE

Motivation

Beaches and sandy coastlines are economically, culturally, and ecologically important, monitoring them is crucial. To understand coastal change, we rely on historical shoreline monitoring data. However, due to the substantial time and resources needed for consistent surveys, only a few locations worldwide have been consistently monitored over the past decades. Satellite-Derived Shorelines (SDS) provide a scalable approach to beach monitoring.



Figure 1: North Shore, O'ahu Coastal Erosion

Satellites

In this work six satellite missions are used: Landsat (5, 7, 8, 9), Sentinel-2, PlanetScope. Landsat imagery covers from 1984 to present day with monthly to bimonthly revisit times and 30 m (Landsat 5) to 15 m (Landsat 7-9) spatial resolution. Sentinel-2 has recorded imagery since June 2015 with 5-day revisit time. The PlanetScope Constellation has recorded the entire land surface of the earth with near-daily revisit time since 2014.

Top of atmosphere radiance values for red, green, blue, near-infrared, and short-wave infrared bands were downloaded. Landsat and Sentinel-2 are publicly available and can be downloaded using the following python package:

<https://github.com/Coastal-Research-Collaborative/geedownload>

PlanetScope imagery is proprietary with free educational licenses available. PlanetScope imagery can be downloaded using the following package (Note PlanetScope does not collect shortwave-infrared data):

<https://github.com/Coastal-Research-Collaborative/planetscope>

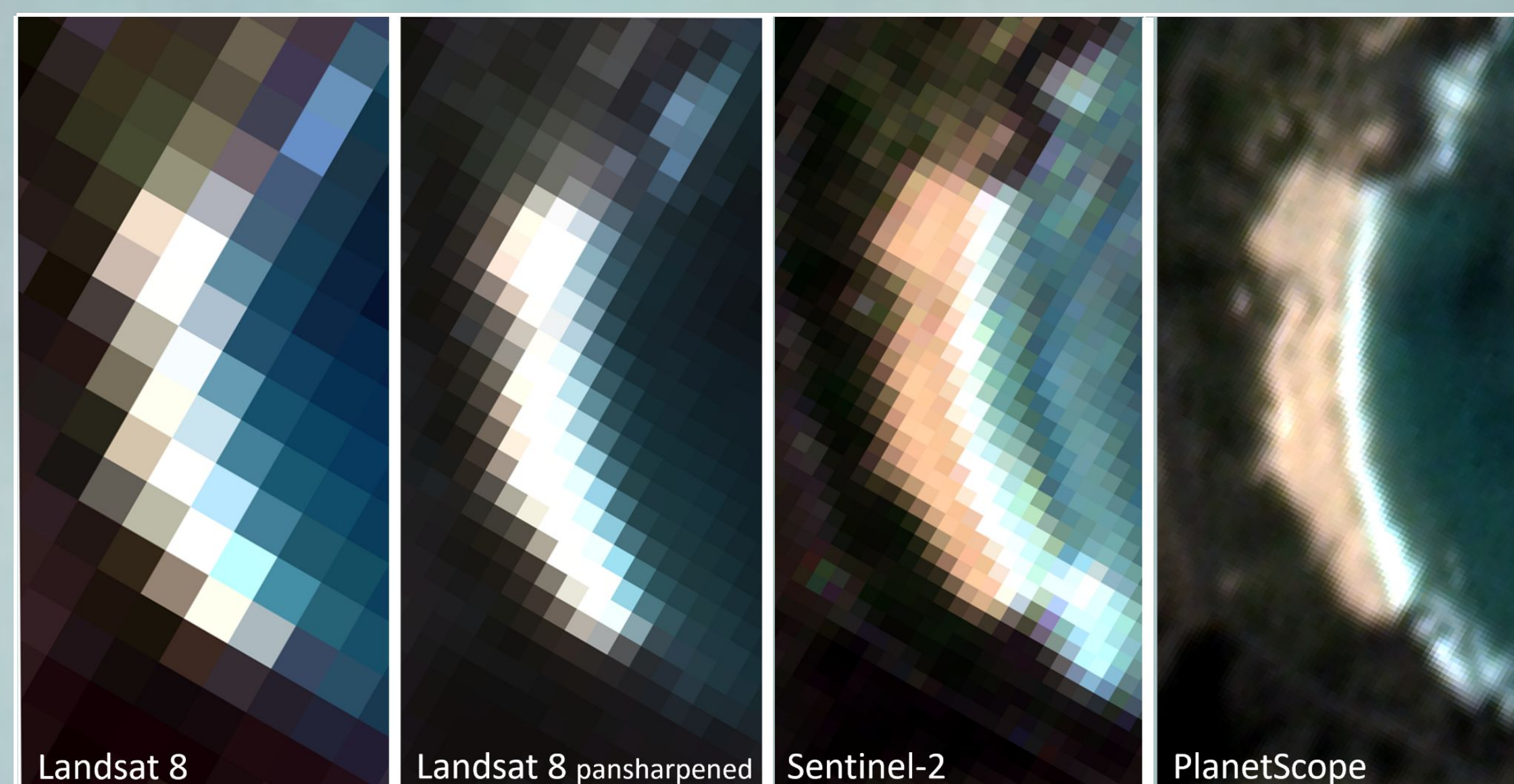


Figure 2: Satellite Sources Makapu'u, O'ahu

Site Description

It is important to validate SDS algorithms against in-situ beach survey data. This study evaluates satellite-derived shorelines against 132 unmanned aerial system (UAS)-derived, manually digitized shoreline surveys collected at 53 beaches on O'ahu between 2010 and the present. The study sites span the west, east, and south shores of the island and encompass a diverse range of coastal morphologies, including reef-fronted shorelines, embayed and headland systems, narrow periodic beaches, and perched beach environments. These sites are predominantly carbonate beach systems subjected to varying wave regimes, providing a representative cross-section of coastal conditions across the island. Results from Landsat 8/9 and Sentinel-2 imagery were evaluated against these UAS observations.



Figure 3: In-situ Site Locations and Transect Positions

Shoreline Extraction

SDS frameworks rely on segmenting the satellite image into land and water and then an algorithm such as the marching squares algorithm is used to draw the shoreline. Traditionally these approaches have mainly been index based, first a gray scale image is created using band math; such as the Normalized Difference Water Index (NDWI) (McFeeters, 1996.). Once the gray scale index image is created a algorithm such as Otsu (Otsu, 1979) is used to define the threshold that best separates land and water. Recently machine learning based image segmentation has become increasingly popular. This includes pixel classifiers such as CoastSat as well as image to image models such as U-Net CNNs.

In this work, two small U-Net models were trained from scratch on 237 variable-sized hand-annotated satellite images from 167 globally distributed beaches (~116 million pixels). One model was trained using standard binary cross-entropy (BCE), while the other incorporated a distance-based weighting that downweights the loss contribution of pixels farther from the shoreline.

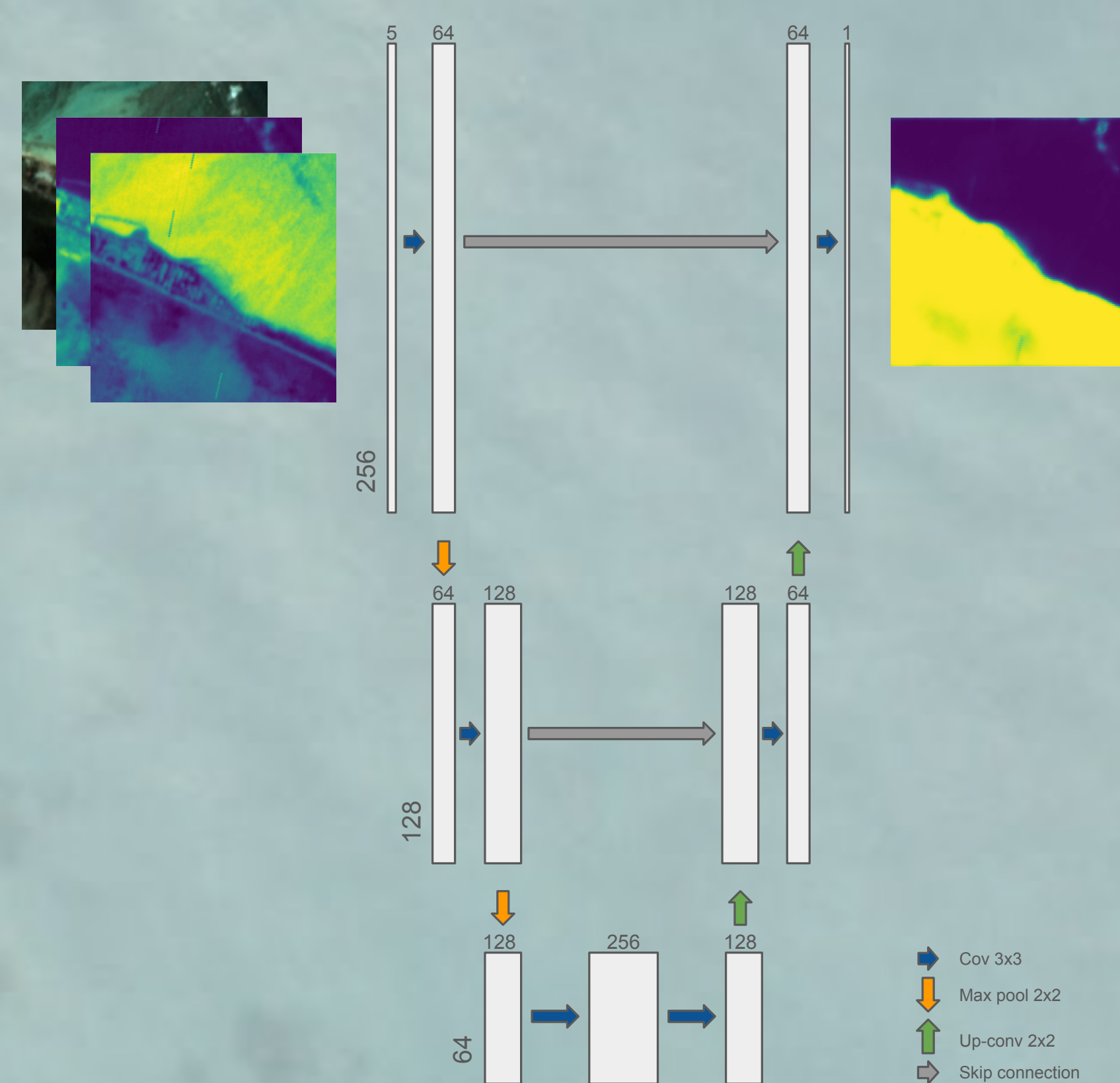


Figure 4: U-Net CNN Architecture

Results

Shoreline positions are traditionally measured along shore normal transects. To compare SDSs and in-situ surveys transect intersections of satellite shorelines within ten days of an in-situ are compared to in-situ intersections. It's important to note that in the current approach there is no correction for how differing tide-level and wave conditions which has been shown to improve performance (Vos et. al, 2023). Figure 5 illustrates this process on a Sentinel-2 image of Sandy Beach, O'ahu.

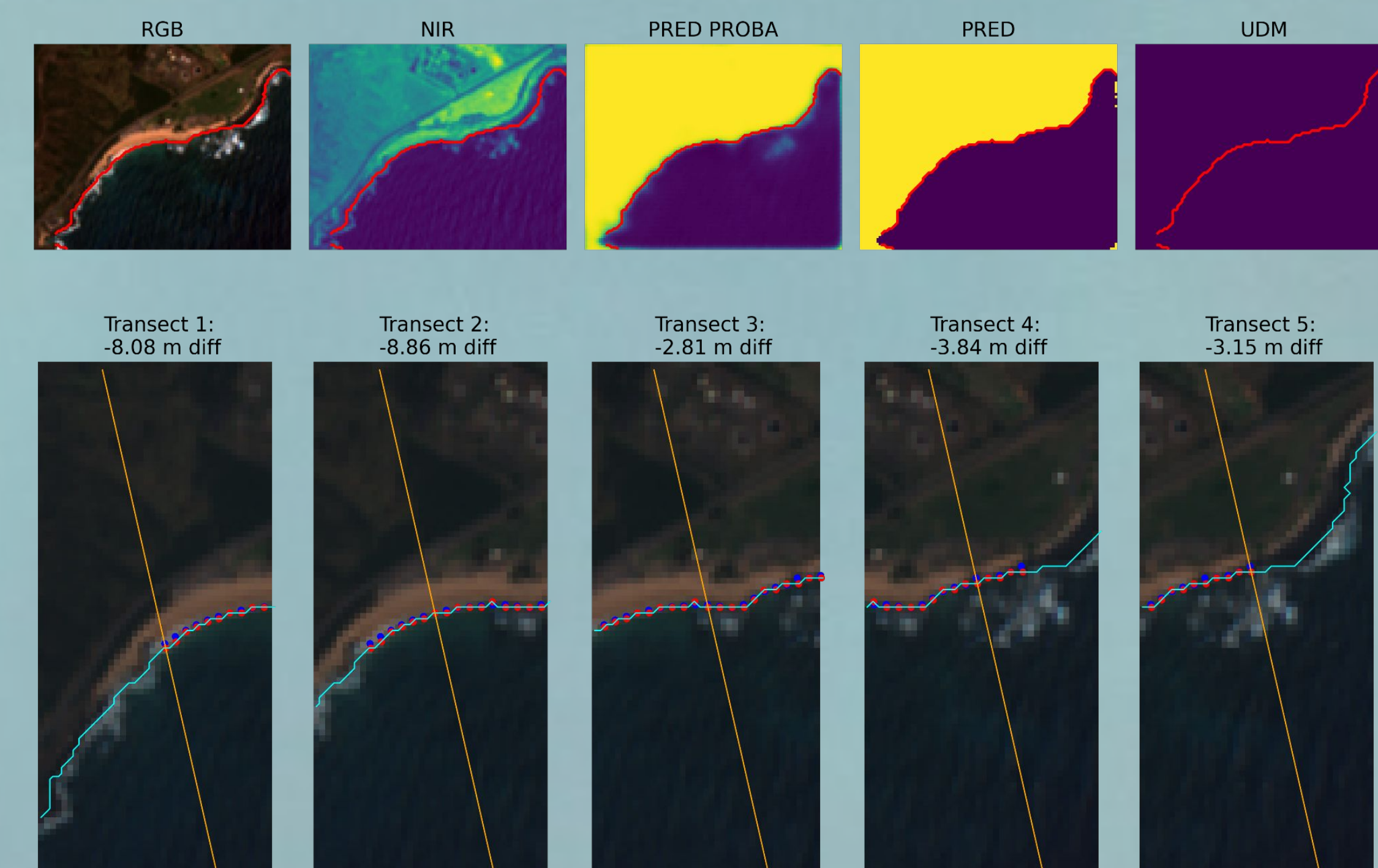


Figure 5: Transect Intersection Comparison Example, Sandy Beach, O'ahu

- Addition of a penalty term to (BCE) upweighting pixels close to the shoreline increases performance
- There are many large errors caused by cloud cover, other artifacts, and catastrophic failures in segmentation

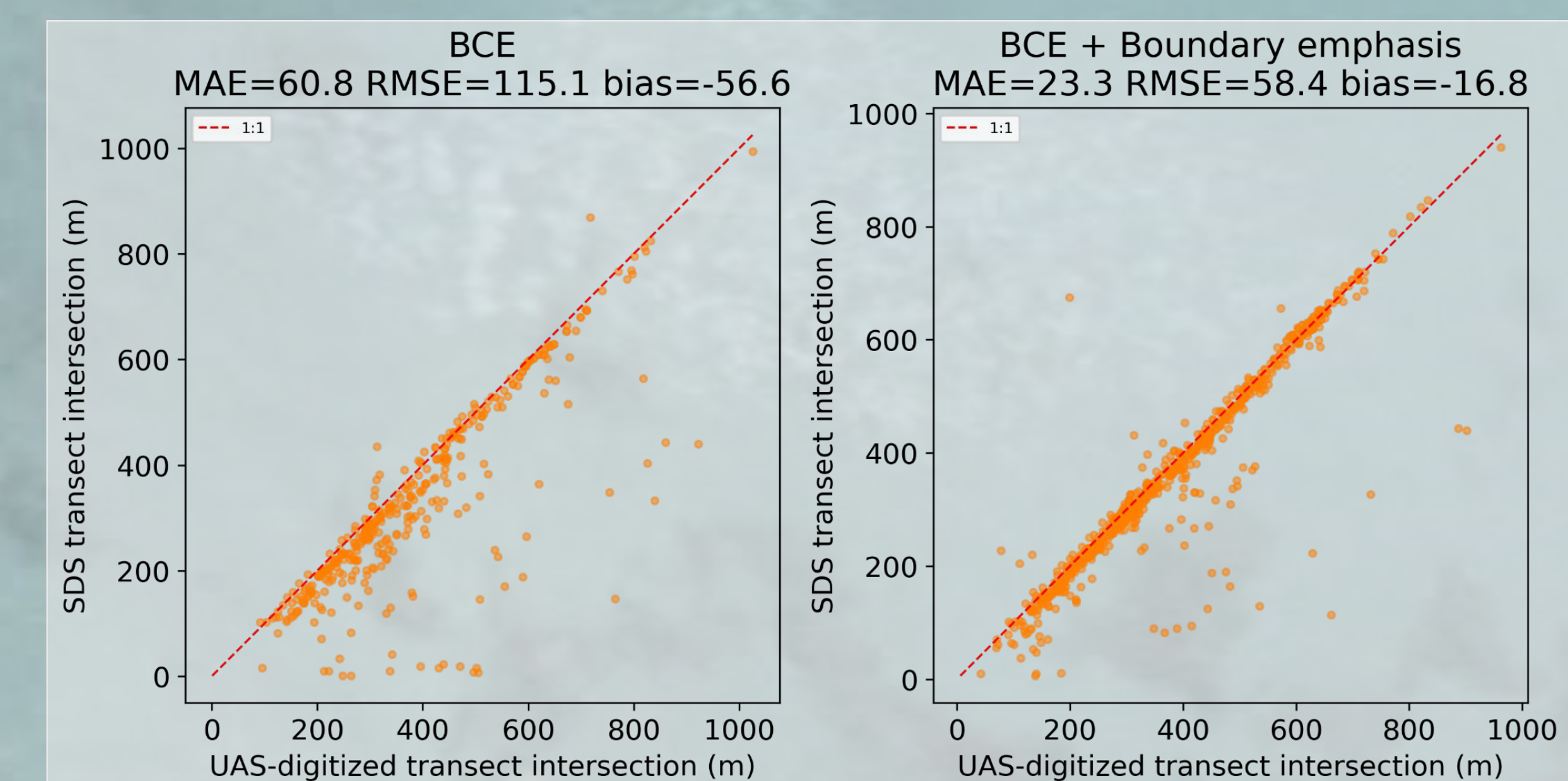


Figure 6: One-to-One Comparison of SDS and UAS-Derived Shoreline Positions Along Transects

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